2026-09-30 | VS&A; DAILY | 10-MINUTE LESSON

Why equal gRMS does not mean equal fatigue damage

Learning objective. Separate input severity from fatigue severity, and recognize the evidence needed before declaring two random-vibration exposures equivalent.

Imagine a test change that preserves 4 gRMS but moves energy toward a bracket resonance. The summary level is unchanged; the bracket sees a different stress history. Today's illustrative calculation turns that familiar warning into a quantitative result: approximately **45 times the predicted damage** for the same exposure duration.

The missing information is frequency placement. For a zero-mean process, squared acceleration RMS is the area under its one-sided power spectral density (PSD). Integrating removes the frequency coordinate. An input dominated by frequencies where the structure is compliant can therefore share an RMS value with one concentrated where the structure transmits little stress.

$$a_{rms}^2 = \int_0^\infty G_a(f)\, df$$
$$G_\sigma(f) = |H_{\sigma a}(f)|^2\, G_a(f)$$

Here, frequency f is in Hz, G_a is in g²/Hz, and the stress frequency-response function $H_{\sigma a}$ is in MPa/g. Its squared magnitude weights every part of the input spectrum before the stress variance is accumulated. This scalar relationship assumes one input and a linear, time-invariant response.

Fatigue adds another weighting: larger stress cycles consume disproportionately more life. The chain to preserve in a review is **input spectrum → structural response → local stress cycles → damage**. A single acceleration number describes only the beginning of that chain.

All spectra, plots and numerical results in this lesson are original illustrative calculations. They are not flight measurements, material allowables or a qualification prescription.

01 / STRUCTURAL FILTERING

Same area, different stress

Consider one elastic mode at 200 Hz with 3% critical damping. Let local stress be proportional to modal relative displacement under base acceleration. Its effective low-frequency stress gain is K = 1 MPa/g; the sign does not affect the PSD.

$$H_{\sigma a}(f) = K / [1 - (f/f_n)^2 + i\, 2\zeta(f/f_n)]$$

Spectrum A is 0.040 g²/Hz from 20 to 420 Hz. Spectrum B is 0.200 g²/Hz from 160 to 240 Hz. Both are zero elsewhere. Their areas are 16 g², so each is 4 gRMS. Both exposures last 120 seconds. The fivefold PSD increase near resonance survives the structural filter; the removed off-resonance energy contributes comparatively little stress.

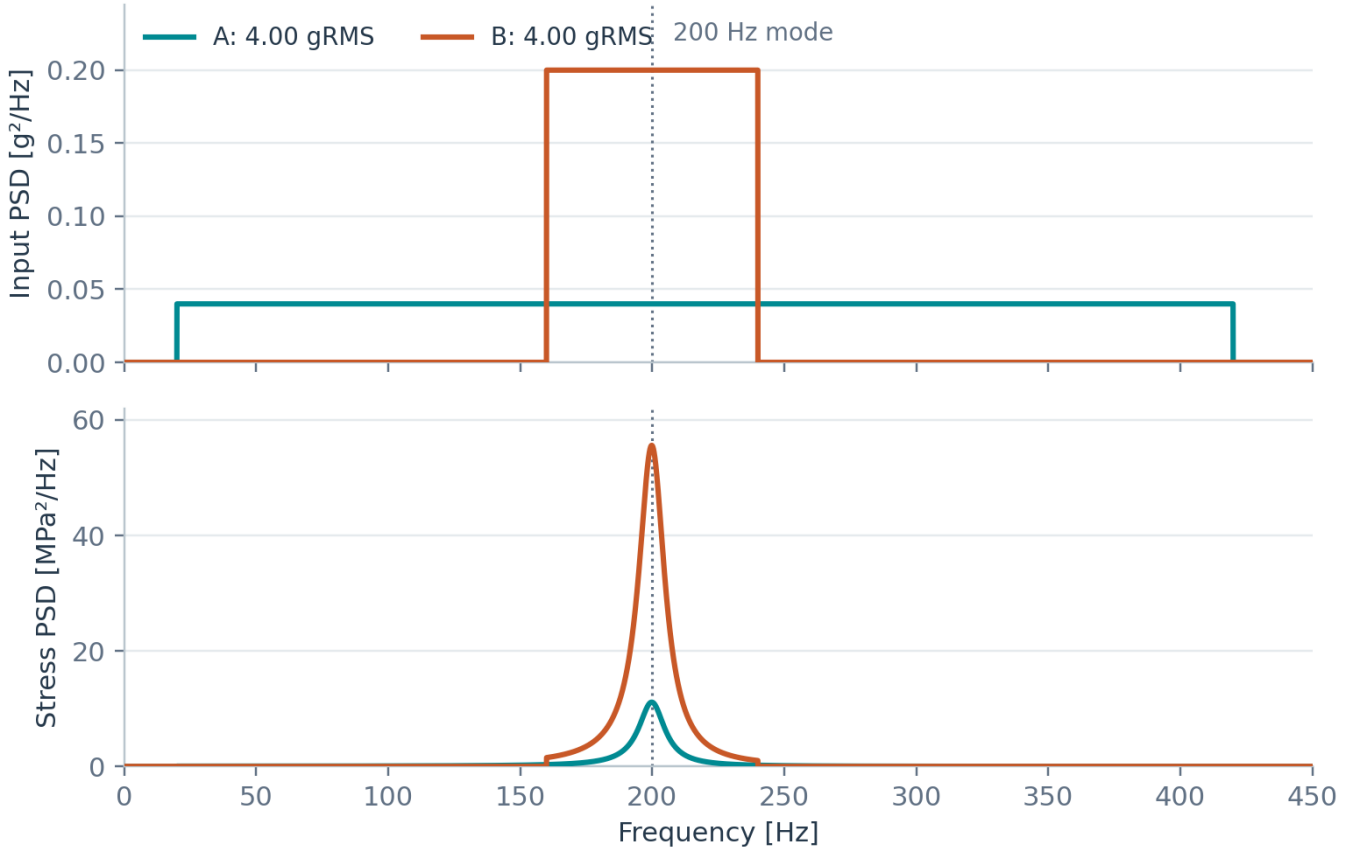


Figure 1. Equal input PSD areas (top) do not produce equal stress PSD areas (bottom). Original linear-model calculation; the dotted line marks the 200 Hz natural frequency. This is a stress/acceleration FRF, not absolute-acceleration transmissibility.

02 / FATIGUE WEIGHTING

From stress variance to damage

For a stationary Gaussian stress process dominated by a narrow frequency band, cycle amplitudes can be approximated by a Rayleigh distribution. Use an amplitude-based S-N law, $N(A)A^m = C$, where A is fully reversed alternating stress amplitude. Combining its amplitude moment with linear Miner accumulation gives the narrow-band estimate below. The spectral method and its limitations are discussed in [1].

$$M_j = \int_0^\infty f^j G_\sigma(f)\, df;\ \sigma_{rms} = \sqrt{M_0};\ \nu_0 = \sqrt{(M_2/M_0)}$$
$$D_{NB} = (T\nu_0/C)(\sqrt{2}\,\sigma_{rms})^m\, \Gamma(1 + m/2)$$

ν_0 is the mean positive-going zero-crossing rate. Moments here use Hz, so no 2π factor is needed. Γ is the gamma function. A stress-range S-N curve requires the corresponding range convention and constant; silently substituting range for amplitude introduces a factor of 2^m.

Illustrative result	Spectrum A	Spectrum B
Input RMS	4.000 g	4.000 g
Stress RMS	14.431 MPa	30.837 MPa
Zero-crossing rate, ν_0	198.390 Hz	199.587 Hz
Bandwidth indicator, α_2	0.9643	0.9934
Normalized damage, m = 5	1.00	44.83

Direct numerical integration gives a stress ratio of 2.1369 and a crossing-rate ratio of 1.0060. For equal duration, material and slope, all other factors cancel:

$$D_B/D_A = (\nu_{0,B}/\nu_{0,A})(\sigma_B/\sigma_A)^5$$
$$= 1.0060 \times 2.1369^5 = \mathbf{44.83}$$

The indicator $\alpha_2 = M_2/\sqrt{(M_0 M_4)}$ approaches one for narrow-band response. These values support using the approximation for teaching, but do not establish its exact rainflow error. No absolute life is claimed: C and actual material behavior have deliberately not been supplied.

03 / SENSITIVITY AND EXPOSURE

A small stress error can become a large life error

The exponent m = 5 is an illustrative choice, not a fitted property of a named alloy or detail. Keep the same spectra and structure but vary that exponent: the calculated damage ratio changes from about 10 to more than 400. This sensitivity makes an unqualified “equal vibration severity” statement particularly risky.

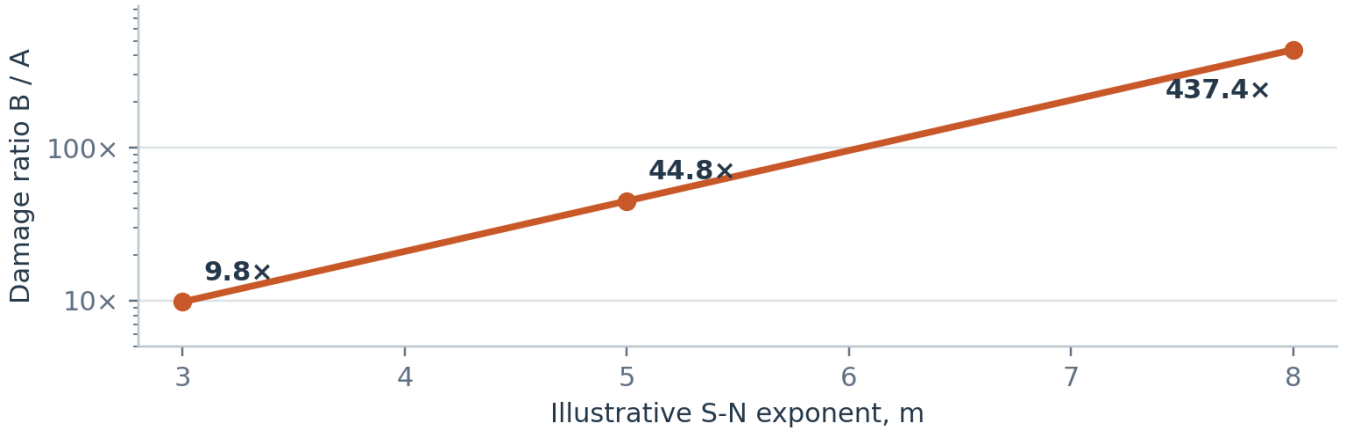


Figure 2. Original sensitivity calculation using the same stress and crossing-rate ratios. Logarithmic vertical axis. The curve shows model sensitivity, not material scatter or measured failure statistics.

A useful reverse check starts with your acceptable damage uncertainty. With m = 5, a 10% stress underestimate corresponds to a true stress 1.10 times the prediction and a damage factor of 1.10^5 = 1.61. Improving an input RMS estimate while leaving the local stress transfer uncertain may therefore address the less consequential uncertainty.

When RMS scaling is useful. If only the level changes while spectrum shape, transfer function and response statistics remain fixed, multiplying the entire PSD by s multiplies stress RMS by √s. Damage then scales as T $s^{m/2}$. Doubling PSD, approximately +3 dB, gives 5.66 times damage for m = 5. Halving duration still leaves 2.83 times damage. Yunis and Ludwiczak discuss this level-duration relationship for specified fatigue-sensitive launch-vehicle structures [2].

That trade is an engineering calculation under stated assumptions. It does not authorize changing a qualification requirement. A test also challenges peak response, workmanship, functional operation and mechanisms that may not follow this S-N law. Preserve those objectives when considering a shorter, higher-level exposure.

04 / ENGINEERING JUDGMENT

What to ask before accepting equivalence

Locate the failure mechanism. Identify the stress quantity, critical detail and load path that control the decision. For a tubing support, a housing foot or a soldered connection, an accelerometer can be a useful witness without being a direct fatigue measurement. Trace the proposed equivalence through the relevant transfer function.

Interrogate spectral placement. Overlay the input spectra with the important mode frequencies and stress-response bands. Repeat the comparison across credible damping, frequency and mounting variation. A nominal mode centered inside an excitation band is a different uncertainty problem from a mode near its edge. Report the resulting damage ratio as a range when the response model warrants one.

Check the response distribution. The narrow-band Gaussian estimate is not a universal conversion from PSD to damage. Multiple separated response peaks require a suitable spectral approximation or time-domain rainflow assessment; Larsen and Irvine compare such methods [1]. Non-Gaussian excitation can also change response statistics. Kihm and colleagues show that kurtosis transfer depends on the excitation and structural filtering [3]. Matching input PSD alone does not establish matching fatigue behavior.

Keep the calculation connected to exposure. Record duration, repeated acceptance runs, screening and operational segments. Segment a nonstationary event where appropriate instead of treating an envelope of unrelated spectral maxima as a measured stationary history. For a synthesized time-history check, vary realizations and record length so one random seed does not masquerade as a reliable life estimate.

Recognize the model boundary. This example has one input, one mode, constant damping, scalar elastic stress and linear damage accumulation. Slip, preload changes, plasticity, mean stress, multiaxial fatigue and load-sequence effects require additional treatment. For multiple correlated inputs, stress PSD depends on the input cross-spectral matrix, including phase and coherence; separate input gRMS values cannot replace it.

In a safety review, ask which failure modes the evidence actually covers. A fatigue calculation can expose an under-test near resonance; it can also expose an unnecessarily severe test that consumes useful life. Neither conclusion follows from gRMS alone. The practical reliability gain comes from preserving the relation between the test environment and the hardware’s failure physics.

Scope reminder. NASA-STD-7001C, dated July 14, 2026, addresses spaceflight payload vibroacoustic environments and tests; its stated scope excludes launch vehicles and several other categories [4]. Do not apply a standard’s levels or margins universally simply because the excitation is called random vibration.

05 / RETAIN AND APPLY

Three takeaways

1. Equal area is not equal response. gRMS discards frequency placement; local stress retains the structure’s frequency weighting.

2. Equal stress RMS is still conditional. Damage also depends on cycle rate, amplitude statistics, duration and the applicable fatigue relation.

3. Equivalence belongs to a failure mechanism. State the response model and uncertainty before using a scalar severity measure to change a test.

Reflection

A revised spectrum keeps gRMS and duration unchanged but doubles the PSD around a narrow, dominant mode. What first-order damage change should you expect for m = 5?

Compact answer. If nearly all damaging stress comes from that band and the transfer function stays linear, stress RMS grows by approximately √2, cycle rate changes little, and damage grows by $2^{5/2} \approx 5.66$. Integrate the actual response spectrum to check the approximation; equal total input area does not cancel the local increase.

Sources and calculation notes

[1] Larsen, C. E., and Irvine, T. (VAL2015). A Review of Spectral Methods for Variable Amplitude Fatigue Prediction and New Results. Equations for spectral moments and narrow-band damage; comparisons with rainflow. The paper identifies VAL2015; NTRS has an inconsistent publication-date field.

[2] Yunis, I., and Ludwiczak, D. (2007). On the Use of 3dB Qualification Margin for Structural Parts on Expendable Launch Vehicles. See Section II.A for fatigue-equivalence scaling. Its recommendations have a restricted hardware scope.

[3] Kihm, F., Rizzi, S. A., Ferguson, N. S., and Halfpenny, A. (July 2013). Understanding How Kurtosis Is Transferred from Input Acceleration to Stress Response and Its Influence on Fatigue Life. RASD 2013.

[4] NASA (July 14, 2026). NASA-STD-7001C: Payload Vibroacoustic Test Criteria. Official standard record and current document; applicability is limited by its scope.

Verified September 30, 2026. Figure and table credit: original calculations prepared for this lesson. Integration used the exact stated FRF over each flat input band. Rounded display values may differ slightly from calculations using full precision.